

Evaluation of a Chromogenic Medium for Enumeration and Differentiation of *Escherichia coli* and Other Coliform Bacteria in Selected Foods

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Introduction

Coliform is a nontaxonomic group of bacteria that are exclusively gram negative, non-spore forming, facultative anaerobic, and rod shaped, and that ferment lactose to produce acid and gas. The group is composed of enteric bacteria such as *Escherichia*, *Citrobacter*, *Klebsiella*, and *Enterobacter*. Enumeration of coliform bacteria may be useful to confirm sanitation of water or food processing environments (Bryant et al. 2003). *Escherichia coli* is found in the intestine of animals and is abundant in human and animal feces (Leclerc and Moriametz 1980). The bacteria can also be isolated from environmental samples (Winfield and Groisman 2003). Fecal coliforms are often used as indicators of contamination and the possible presence of pathogenic organisms. The fecal coliform group consists primarily of *E. coli* and some thermotolerant *Klebsiella*. While these bacteria are generally nonpathogenic, they can cause infection in immunocompromised hosts (Nataro and Kaper 1998).

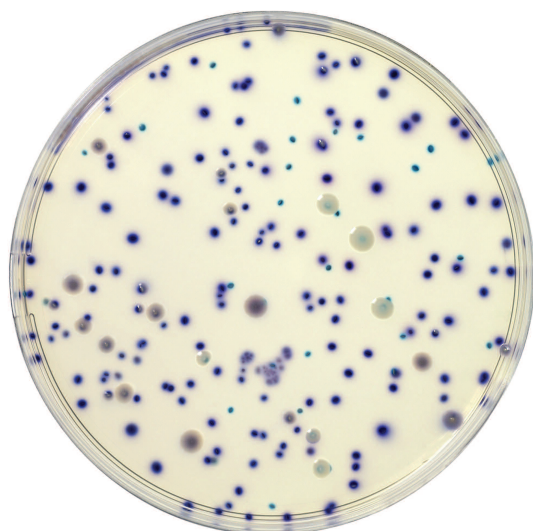


Fig. 1. RAPID'E.coli 2 agar plate. Coliform bacteria other than *E. coli* produce blue-green colonies, while *E. coli* produce violet-pink colonies.

Most probable number (MPN) methods for enumeration of coliform bacteria can be laborious and costly. The use of chromogenic substrates in media has led to the development of faster and easier methods for detection, differentiation, and enumeration of target bacteria (Manafi 2000).

RAPID'E.coli 2™ agar is a chromogenic medium for direct enumeration of coliforms, including *E. coli*, in food. This medium contains chromogenic substrates to simultaneously detect the activity of two enzymes, β -D-glucuronidase (GLUC) and β -D-galactosidase (GAL). The GAL-specific substrate results in blue-green coloration of colonies positive for GAL, while the GLUC-specific substrate results in violet coloration of colonies positive for GLUC. Coliforms other than *E. coli* are GAL positive and GLUC negative (GAL+/GLUC-) and form blue to green colonies, while *E. coli* have both enzymes (GAL+/GLUC+) and form violet colonies (Figure 1). A count of total coliforms can be obtained by adding the number of blue-green colonies and the number of violet colonies.

This report summarizes a method comparison study (AOAC Research Institute Validation Study). In this study, RAPID'E.coli 2 medium was granted Performance Tested Method status by AOAC-RI, certificate #050601.



Methods

Sample Preparation

Method comparison studies were performed to compare the RAPID'E.coli 2 agar method to AOAC official method 966.24, the reference method, for enumeration of *E. coli* and other coliforms in food matrices (Figure 2). The following 11 foods were selected: raw ground beef, raw boneless pork, fermented sausage, processed ham, processed turkey, frozen turkey breast, raw ground chicken, cottage cheese, processed ricotta cheese, unpasteurized raw milk, and dry infant formula. Food was purchased at a local grocery store and screened for the presence of naturally occurring *E. coli* or other coliform bacteria. Three levels of each organism were targeted; 10–100, 100–1,000, and 1,000–10,000 colony forming units per gram (cfu/g). If available, naturally contaminated products

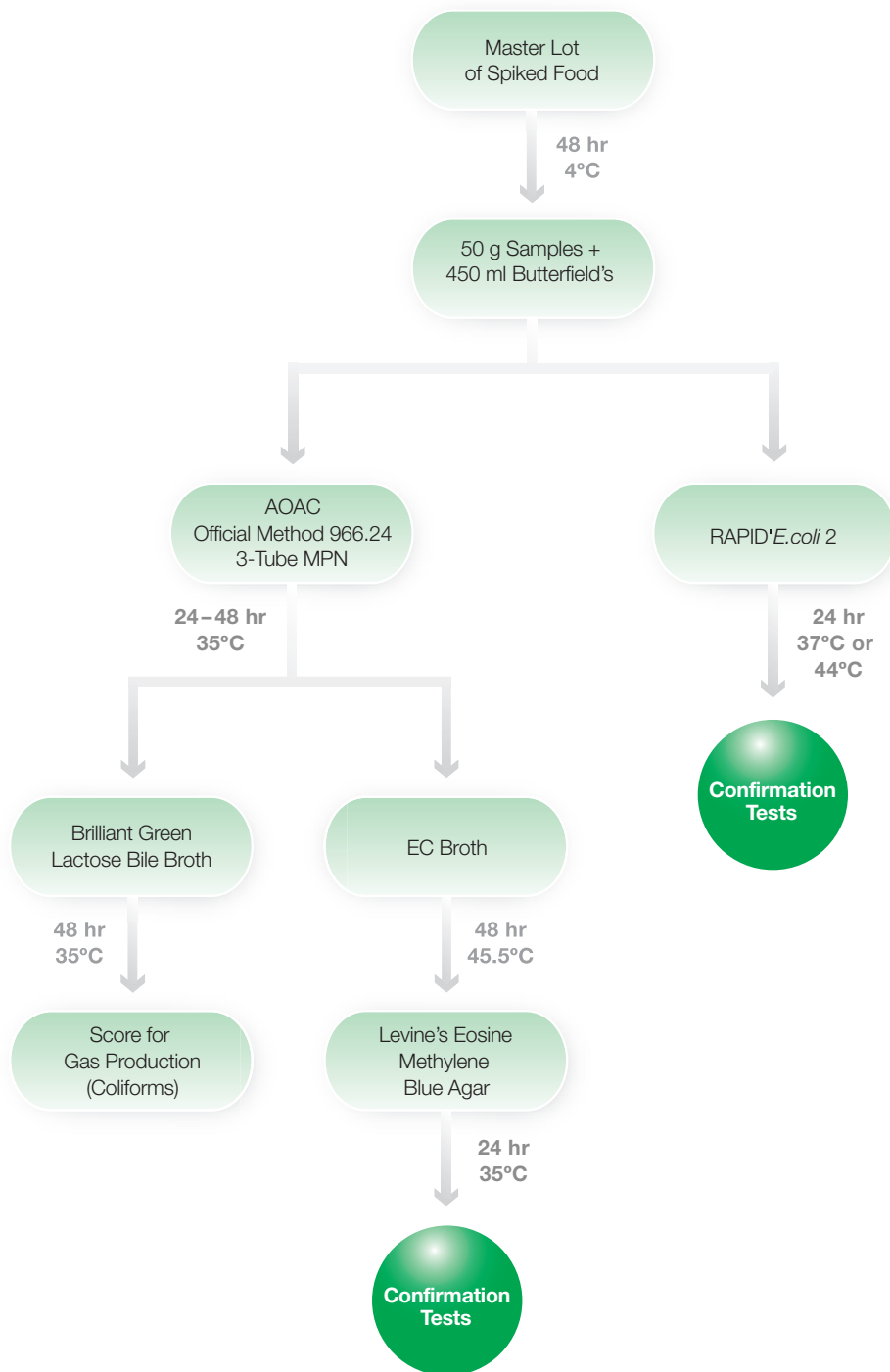


Fig. 2. Flowchart of study design.

were used for analysis. If naturally contaminated food was not available, one strain of *E. coli* and one non-*E. coli* coliform strain were inoculated into each food. For some foods, naturally contaminated levels were supplemented by inoculation to obtain desired target levels (Table 1). Uncontaminated controls were not examined if only contaminated lots were obtained.

Table 1. Inoculum cultures used for different food types. SLR refers to isolates from the culture collection of Silliker Laboratories (South Holland, IL, USA).

Food	<i>E. coli</i>	Other Coliform
Raw ground beef	SLR 555	<i>E. aerogenes</i> ATCC 13048
Raw boneless pork	SLR 651	<i>C. freundii</i> ATCC 6879
Fermented sausage	ATCC 25922	<i>K. pneumoniae</i> ATCC 13883
Processed ham	ATCC 51813	<i>E. cloacae</i> ATCC 13047
Processed turkey	ATCC 13762	<i>K. oxytoca</i> ATCC 43863
Frozen turkey breast	SLR 650	<i>C. freundii</i> SLR 788
Raw ground chicken	ATCC 35218	<i>E. aerogenes</i> ATCC 13408
Cottage cheese	ATCC 11775	<i>E. cloacae</i> ATCC 7256
Processed ricotta cheese	SLR 636	<i>C. freundii</i> SLR 2155
Raw milk	Naturally contaminated	Naturally contaminated
Dry infant formula	ATCC 11775	<i>C. freundii</i> ATCC 8090

For supplementing foods that did not meet target levels of contamination, inoculated cultures were individually grown in BD BBL trypticase soy broth (TSB, Becton, Dickinson and Co.) at 35°C for 18–24 hr. Dilutions were prepared with Butterfield's phosphate buffer (0.3 mM, pH 7.2). The number of colony forming units was determined by plating serial dilutions on trypticase soy agar (TSA, BD BBL), incubating at 35°C for 18–24 hr, and then counting colonies. Broth cultures were stored at 4°C prior to food inoculation. Food products were then inoculated to the appropriate target level. Following inoculation, products were held at 4°C for 48 hr before analysis to allow the bacteria to equilibrate to the food environment.

Five 50 g samples of each food at each contamination level were homogenized by blending with 450 ml Butterfield's phosphate buffer for 2 min. Two subsamples of each of the five homogenates were prepared for analysis: one by the reference method and one by the RAPID'*E. coli* 2 method.

AOAC Official Method 966.24

Reference method samples were processed by a three-tube MPN method. Specifically, 1 ml of each homogenized sample was inoculated into lauryl sulfate tryptose (LST, BD BBL) broth at 1:10, 1:100, and 1:1,000 dilutions, with triplicate tubes at each dilution. Tubes were incubated at 35°C and scored for gas formation at 24 hr and 48 hr. Using a 3 mm loop, samples from gassing tubes were transferred to Brilliant Green lactose bile (BGLB, BD BBL) broth and *E. coli* (EC, BD BBL) broth. BGLB broth tubes were incubated at 35°C for 48 ± 2 hr. The most probable number of total coliform bacteria per gram was computed by scoring the number of gas-positive BGLB tubes at each dilution and referencing MPN Table 966.24A. EC broth tubes were incubated at 45.5°C for 48 ± 2 hr. Gas-positive tubes were streaked to Levine's eosine Methylene Blue (LEMB,

BD BBL) agar and incubated at 35°C for 24 ± 2 hr. A minimum of two typical colonies with dark purple center and green metallic sheen were selected from each LEMB plate for confirmation. A Gram stain was performed; LST broth was inoculated, incubated at 35°C for 48 ± 2 hr, and scored for gassing; and IMViC biochemicals (indole, Methyl Red, Vogues-Proskauer, citrate) were inoculated. Confirmed *E. coli* isolates were gram negative, exhibited growth and gas in LST, and were indole- and Methyl Red-positive and Vogues-Proskauer- and citrate-negative.

RAPID'*E. coli* 2 Method

For samples analyzed according to the RAPID'*E. coli* 2 method, serial dilutions, determined by target inoculation level, were made in Butterfield's phosphate buffer. Two 1 ml aliquots were removed from each sample and placed in separate empty petri dishes. Tempered RAPID'*E. coli* 2 agar (10 ml) was added to each petri dish, and samples were swirled to mix. One plate was incubated at 37°C, the other at 44°C. Blue colonies were counted as suspect non-*E. coli* coliforms, including fecal coliforms. Coliform isolates were confirmed by picking two suspect colonies and subculturing to LST and BGLB broths, as described above. Colonies producing growth and gas in LST and BGLB were considered confirmed for non-*E. coli* coliforms. The number of non-*E. coli* coliforms was determined by multiplying the percent of colonies confirmed as positive by the total number counted using the RAPID'*E. coli* 2 method. The confirmed number of colonies was then multiplied by the appropriate dilution factor.

Violet colonies on RAPID'*E. coli* 2 agar were counted as suspect *E. coli*. Two suspect *E. coli* isolates were confirmed by performing Gram staining, subculturing to LST broth and LEMB agar, and completing IMViC biochemical tests, as described above. The total number of *E. coli* colonies was determined by multiplying the percent of isolates confirmed as positive by the total number of violet colonies counted using the RAPID'*E. coli* 2 method. The confirmed number of *E. coli* was then multiplied by the appropriate dilution factor. Since *E. coli* is a coliform, enumeration of total coliforms was achieved by adding of the number of non-*E. coli* coliforms to the number of *E. coli*.

Statistical Analysis

Analysis of variance (paired Student's *t*-test) was performed on the results from each food to determine the statistical difference between the RAPID'*E. coli* 2 method (at each temperature) and the AOAC method. If there was a missing value, the missing value and the corresponding value for the other method were removed from analysis.

Results

Using paired *t*-tests, no statistical differences were observed between the RAPID'*E. coli* 2 method and the AOAC method for raw ground beef, processed ham, processed turkey, and frozen turkey breast (Table 2). Few statistical differences were observed between the two methods for raw boneless pork,

Table 2. Analysis of variance. A p-value of <0.05 indicates a statistical difference. A result of NA indicates one of the methods did not produce enough data points for the comparison, since the points must be paired for this test.

Food	Level	p-Value 37°C		p-Value 44°C	
		<i>E. coli</i>	Coliform	<i>E. coli</i>	Coliform
Raw ground beef	Low	NA	0.88	0.44	0.44
	Mid	0.77	0.21	0.66	0.68
	High	0.18	0.26	0.28	0.10
Raw boneless pork	Low	0.73	0.68	0.77	0.14
	Mid	0.43	0.31	0.66	0.05
	High	0.001	NA	0.08	NA
Fermented sausage	Low	0.02	0.07	0.01	0.50
	Mid	0.22	0.87	0.15	0.15
	High	0.10	0.46	0.42	0.26
Processed ham	Low	0.97	0.80	0.24	0.36
	Mid	0.25	0.07	0.24	0.13
	High	0.85	0.06	0.69	0.12
Processed turkey	Low	0.88	0.34	0.57	0.16
	Mid	0.59	0.39	0.46	0.30
	High	0.38	0.48	0.14	0.97
Frozen turkey breast	Low	0.25	0.94	0.14	0.09
	Mid	0.63	0.09	0.77	0.06
	High	0.38	0.78	0.80	0.72
Raw ground chicken	Low	0.80	0.01	0.39	0.78
	Mid	0.03	0.00	0.11	0.01
	High	0.07	0.15	0.05	0.02
Cottage cheese	Low	0.09	0.85	0.0006	0.001
	Mid	0.08	0.50	0.07	0.004
	High	0.36	0.99	0.27	0.13
Processed ricotta cheese	Low	0.70	0.17	NA	NA
	Mid	0.17	0.28	0.11	0.11
	High	0.13	0.19	0.07	0.03
Raw milk	Low	0.01	0.84	0.17	0.68
	Mid	0.79	0.91	0.73	0.12
	High	0.14	0.60	0.30	0.17
Dry infant formula	Low	0.00	0.02	0.00	0.87
	Mid	0.02	0.15	0.03	0.43
	High	0.16	0.53	0.02	0.54

fermented sausage, raw ground chicken, raw milk, and dry infant formula. For these foods, the RAPID'*E. coli* 2 method produced higher counts, consistent with the range of the target inoculum. This suggests that for these foods, the RAPID'*E. coli* 2 method yielded more accurate results than the reference method. Cottage cheese and processed ricotta cheese produced optimal results at 37°C only.

Discussion

The data presented in this report represent the method comparison portion of the Performance Tested Method AOAC Research Institute Validation Study of RAPID'*E. coli* 2, in which the AOAC official method was the appointed reference method for comparison. The RAPID'*E. coli* 2 method was shown to be effective for enumerating *E. coli* in food and was granted Performance Tested Method status.

We believe many of the statistical differences observed between the MPN method and the RAPID'*E. coli* 2 method can be attributed to method differences. MPN analysis is based on statistical probability in a mixed liquid culture, whereas the direct plating method is based on enumeration of individual colonies grown from viable cells on solid medium, which can be selected off of the plate and confirmed. The latter method is expected to be more accurate since it involves direct counting rather than probabilities. Indeed, we found that where statistical differences occurred between the two methods, results obtained from RAPID'*E. coli* 2 more accurately reflected the levels inoculated into the food products.

For most of the foods tested, MPN results yielded lower counts than the RAPID'*E. coli* 2 method. Nonetheless, the RAPID'*E. coli* 2 results were consistent at both incubation temperatures, suggesting better recovery with the RAPID'*E. coli* 2 method. The RAPID'*E. coli* 2 results also were more consistent with the target inoculum levels than the MPN results.

Conclusions

RAPID'*E. coli* 2 agar allows enumeration and differentiation of *E. coli* and other coliform bacteria in 24 hr. When compared to AOAC official method 966.24, RAPID'*E. coli* 2 agar was an effective and efficient method for enumeration of coliforms and *E. coli* in raw ground beef, raw boneless pork, fermented sausage, processed ham, processed turkey, frozen turkey breast, raw ground chicken, cottage cheese, processed cheese, raw milk, and dry infant formula. Furthermore, the RAPID'*E. coli* 2 method greatly reduced the time to results (by days) compared to the MPN method. In addition, multiple tubes of media were not required to process samples.

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